Driver Drowsiness Detection using Deep Learning

Based on a Custom CNN Architecture

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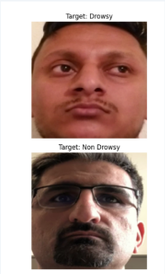
MOTIVATION

Driver fatigue is a major cause of road accidents globally. Our goal is to develop a real-time, accurate system to detect drowsiness based on visual facial cues (specifically eye closure) to prevent fatal accidents.

THE DATASET (DDD)

We utilized the Kaggle Driver Drowsiness Dataset (DDD):

- **Size:** ~41,000 images
- **Classes:** Drowsy (Closed) vs. Non-Drowsy (Open)
- **Split:** 60% Train, 20% Val, 20% Test



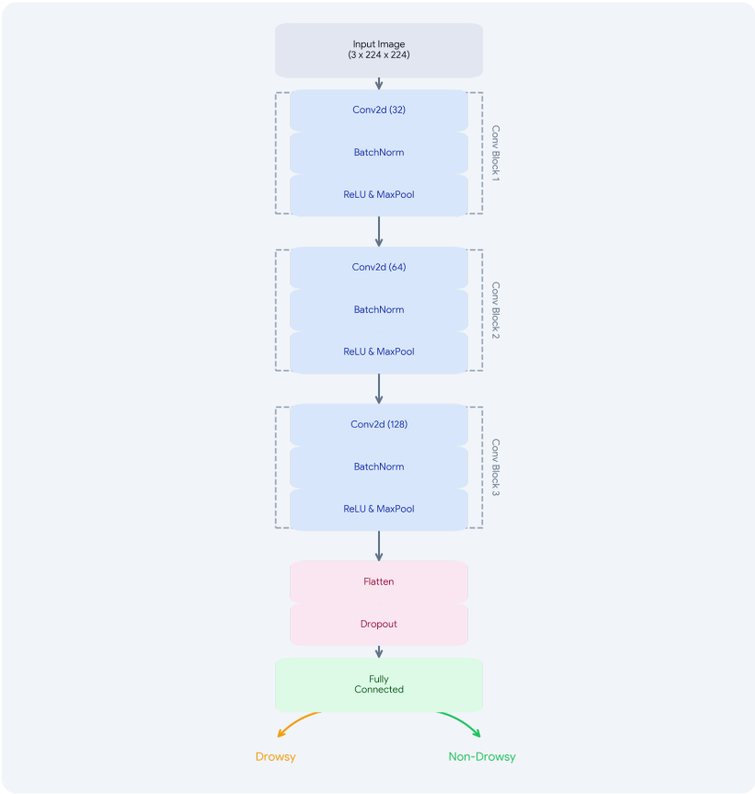
DATA PREPARATION

- **Preprocessing:** Images resized to 224x224 and normalized to PyTorch tensors.
- **Augmentation:** Applied Random Rotation and Color Jitter to enhance robustness.

PROPOSED CNN ARCHITECTURE

We built a custom Convolutional Neural Network from scratch, optimized for fast convergence and high spatial feature extraction.

- **3 Convolutional Blocks:** Conv2d, BatchNorm, ReLU, MaxPool.
- **Regularization:** Dropout layers to prevent overfitting.

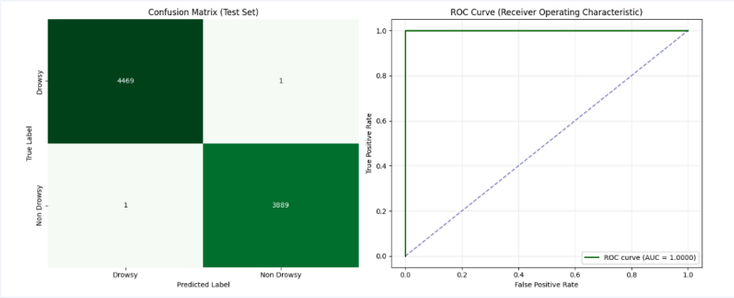


EXPERIMENTAL RESULTS

Test Accuracy: 99.9%

AUC Score: 1.0000

Only 2 misclassifications out of 8,360 samples!



CONCLUSION & FUTURE WORK

The custom CNN model demonstrated superior performance, proving that the spatial features of eye closure are highly discriminative.

Future Directions:

- **Video Analysis:** LSTMs or Transformers to analyze sequences.
- **Edge Deployment:** Optimizing via ONNX for mobile platforms.